AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a.

i. Simplify $\frac{(x+y)^3}{x^2-y^2} \div \frac{(x+y)^2}{x^4-y^4}$ to its simplest form. (3 Marks)

ii. If $x = \sqrt{2} - 1$, then express $\frac{1}{x}$ in the simplest rationalised form. (2 Marks)

b. If $(a+b)^2 = 49$ and $a^2 + b^2 = 25$, then find the value of $a - b$. (5 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 5 Marks)

a. Completly facotrise $3a^2 - 3 - (a-1)^2$. (5 Marks)

b. Factorise the following expressions completely.

i. $6x^2 - x - 2$ (3 Marks)

ii. $8x^3 - 27$ (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. Find the highest common factor (H.C.F.) of the following expressions.

$y^2 - b^2$, $(y+b)^2$ and $y^3 - b^3$

b. Write the following algebraic expression in its simplest form.

$$\left(1 - \frac{x^2 + y^2}{x^2 - y^2}\right) + \left(\frac{y^4 + 1}{x^2 - y^2}\right)$$

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Q.4. (Total 5 Marks)

i. Find the value of x for $2x+\frac{x}{3}=-7$, where $x \in z$. (3 Marks)

ii. Find the value(s) of x for $|x-18|=18$, where $x \in z$. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 5 Marks)

a. Find the solution set of $2x^2 - 3x - 2 = 0$.

b. The product of two consecutive positive even integers is 168. Find the larger integer of two by using algebraic method.

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. Do as directed.

i. Write an example of rectangular matrix. (1 Mark)

ii. Write an example of scalar matrix. (1 Mark)

iii. Write a matrix of order 2×3. (1 Mark)

iv. Find the transpose of the matrix $\begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$. (1 Mark)

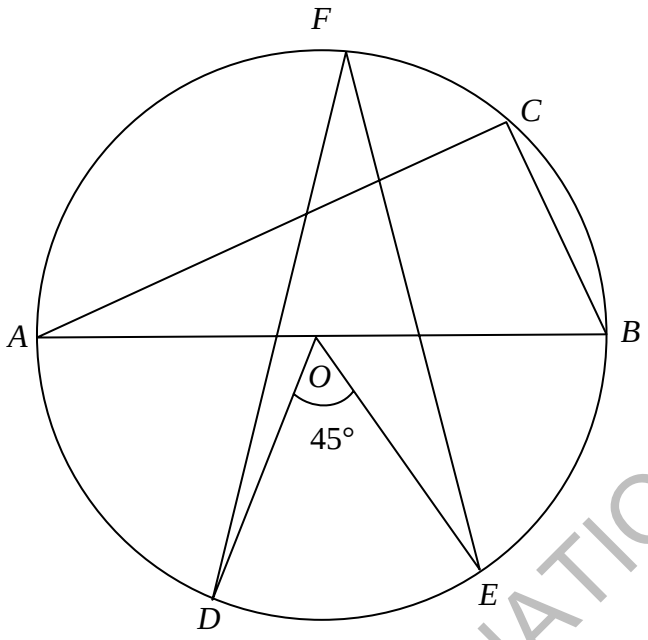
v. Find the additive inverse of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. (1 Mark)

b. For $\begin{matrix} x + y = 5 \\ x - y = 1 \end{matrix}$, find the values of *x* and *y* by using Cramer's rule. (5 Marks)

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Q.7. (Total 5 Marks)

In the given figure, *O* is the centre of the circle and *AB* is the diameter.



i. Find $m\angle DFE$ and write the reason to justify your answer. (2 Marks)

ii. Find $m\angle ACB$ and write the reason to justify your answer. (2 Marks)

iii. Identify the obtuse angle in the given figure, if any. (1 Mark)

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Q.8. (Total 4 Marks)

In the given space, draw a rectangle whose one of the side is 6 cm and diagonal is 10 cm long.

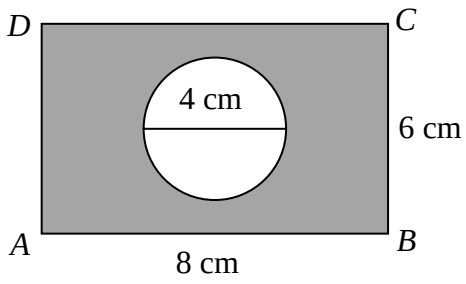
Space for diagram

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Q.9. (Total 4 Marks)

In the given diagram, find the area of the shaded region.

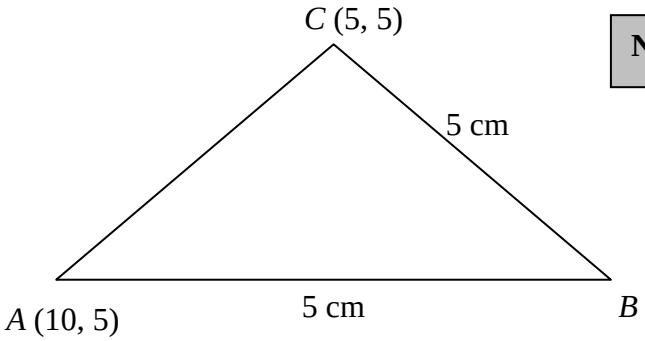
Note: $\pi = 3.14$ approx.



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Q.10. (Total 3 Marks)

Verify that the given triangle *ABC* is an equilateral triangle.



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END OF PAPER

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